Fixed Income

Bond markets, valuation, risk & credit analysis

Bond Features & Types

LM1–3 · Instrument design, cash flows, issuance

BOND ANATOMY: KEY FEATURES

Structural Elements

- Par (face) value — repaid at maturity
- Coupon rate — % of par paid periodically
- Maturity date — when principal is due
- Issue / clean price at issuance

Coupon Types

- Fixed-rate: same coupon every period
- Floating-rate: reference + spread (e.g. SOFR+150bp)
- Zero-coupon: no coupons, deep discount
- Step-up: coupon rises over time

Embedded Options

- Callable: issuer may redeem early
- Putable: holder may sell back
- Convertible: exchange for equity
- Options affect yield & duration

A bond is a promise to pay coupons + par. Each feature directly affects valuation and risk.

BOND TYPES BY ISSUER & STRUCTURE

Type	Issuer	Key Feature	Risk Level
Government Bond	Sovereign / Treasury	AAA benchmark, liquid	Low
Agency Bond	Govt-sponsored entity	Implicit/explicit guarantee	Low–Med
Municipal Bond	State / city	Often tax-exempt income	Low–Med
Corporate Bond	Public company	Credit spread over govt	Med–High
High Yield Bond	Sub-investment grade co.	Higher coupon, higher risk	High
Covered Bond	Bank	Dual recourse to issuer + pool	Low–Med
ABS / MBS	SPV	Backed by asset pool	Variable

BOND ISSUANCE & MARKET STRUCTURE

Primary Market

- Public offering via syndicate (bookbuilding)
- Private placement: direct to investors
- Auction: governments sell T-bills/bonds
- Shelf registration: pre-approved issuance

Secondary Market

- OTC dealer market: most bonds trade
- Exchange-traded: smaller corporate issues
- Repo market: short-term secured financing
- Bid-ask spread = dealer's compensation

Bond Indenture

- Legal contract: trustee protects holders
- Affirmative covenants: must maintain
- Negative covenants: must not do
- Cross-default: linked to other obligations

Short-Term Instruments

- Commercial paper: < 270 days, no coupons
- Banker's acceptances: trade finance
- Certificates of deposit (CD)
- Repurchase agreements (repos)

Bond Valuation

LM4 · Pricing, yield measures & clean/dirty price

BOND PRICING: PRESENT VALUE OF CASH FLOWS

$$P = C/(1+r)^1 + C/(1+r)^2 + \ldots + (C+FV)/(1+r)^n$$

At Par

- Coupon rate = YTM
- Price = 100
- Bond priced at face value
- Yield = quoted coupon

At Discount

- YTM > coupon rate
- Price < 100 (par)
- Investor gains capital
- Higher required return

At Premium

- YTM < coupon rate
- Price > 100 (par)
- Investor loses capital
- Lower required return

YIELD MEASURES

Current Yield

- CY = Annual Coupon / Price
- Ignores time value and capital gain
- Quick approximation only

Yield to Maturity (YTM)

- IRR of buying bond at P and holding to maturity
- Assumes coupons reinvested at YTM
- Most widely quoted yield measure

Yield to Call (YTC)

- IRR assuming bond called at first call date
- Relevant when bond trades at premium
- YTC < YTM if premium bond is callable

Yield Spread Measures

- G-spread: vs govt benchmark
- I-spread: vs swap rate curve
- Z-spread: constant OAS vs spot curve
- OAS: after stripping embedded option

YTM implicitly assumes reinvestment at YTM — violated in practice. Use horizon return for more realistic analysis.

CLEAN PRICE, FULL PRICE & ACCRUED INTEREST

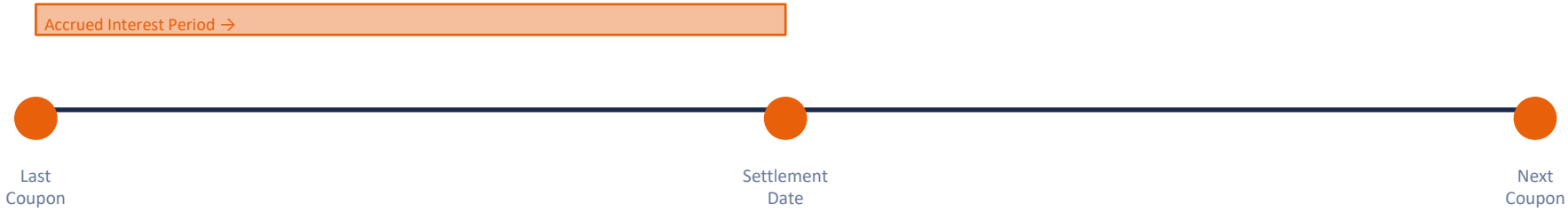
Full Price (Dirty) = Clean Price + Accrued Interest | AI = Coupon × (Days since last coupon / Days in period)

Clean Price (Flat Price)

- Quoted price in the market
- Excludes accrued interest
- Allows comparison across settlement dates
- Does NOT change on coupon date itself

Full Price (Invoice / Dirty)

- Actual cash paid to seller
- Includes accrued interest
- Clean price + AI = full price
- Drops sharply on ex-coupon date



MATRIX PRICING & INTERPOLATION

What is Matrix Pricing?

- Used when bond lacks recent market trade
- Estimate YTM by interpolating between comparable bonds
- Comparable bonds: similar issuer, maturity, credit rating
- Then back-solve for estimated price

Steps

- 1. Identify nearest comparable bonds (above & below maturity)
- 2. Interpolate YTM proportionally by maturity gap
- 3. Calculate estimated price using interpolated YTM
- 4. Adjust for any differences in coupon, credit, liquidity

Bond	Maturity	YTM
Bond A	3 yr	4.20%
Target	5 yr	?
Bond B	7 yr	5.60%

$$YTM^* \approx 4.20 + (5.60-4.20) \times 2/4 = 4.90\%$$

Linear interpolation between 3yr and 7yr benchmarks

Spot & Forward Rates

LM9 · Zero rates, bootstrapping & forward curve

SPOT RATE CURVE & BOOTSTRAPPING

Spot Rate (Zero Rate)

- Rate for a single cash flow at time t
- Derived from zero-coupon bonds
- S_1, S_2, S_3 ... = spot rate curve
- Used to price any cash flow stream precisely

$$P = CF_1/(1+S_1)^1 + CF_2/(1+S_2)^2 + \dots + CF_n/(1+S_n)^n$$

Key Insight

- Bootstrapping extracts spot rates from coupon bond prices
- Each new spot rate solved sequentially using known earlier spots

Bootstrapping Example

Year	Par Bond YTM	Spot Rate
1	3.00%	3.00%
2	3.50%	3.52%
3	4.00%	4.04%
4	4.40%	4.46%

Spot rates exceed par rates when curve is upward-sloping

FORWARD RATES

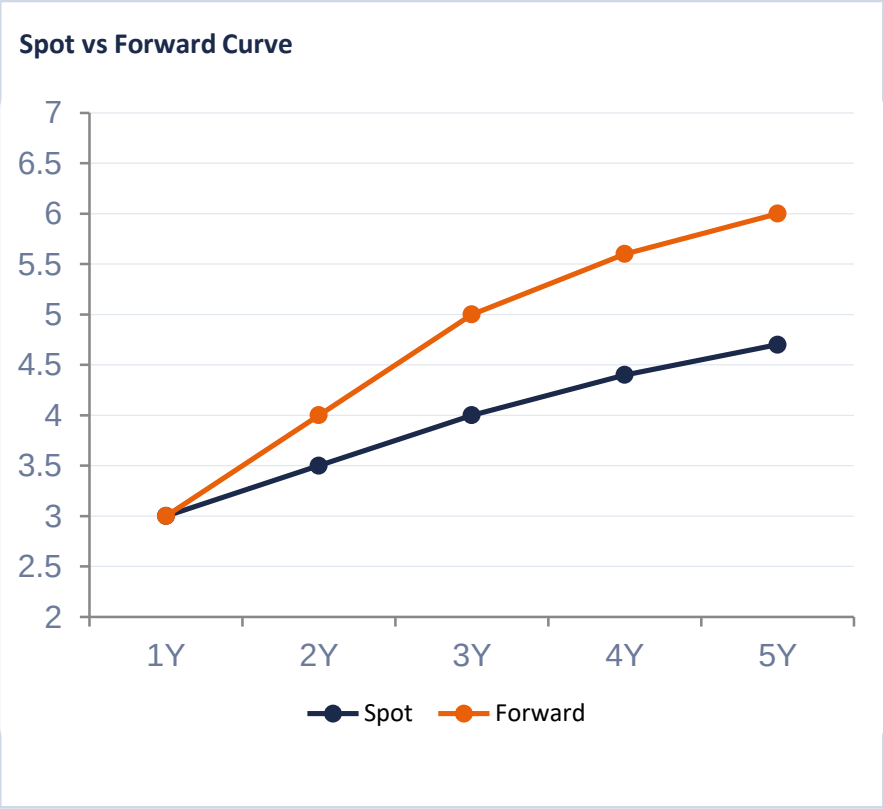
What Is a Forward Rate?

- Rate agreed today for lending/borrowing in the future
- Notation: f(T*, T) = forward for T years starting in T*
- Derived from spot rate curve (no-arbitrage)
- Forward curve > spot curve when spot is rising

$$(1+S_2)^2 = (1+S_1)^1 \times (1+f(1,1))^1$$

Example

- S_1=3%, S_2=4% → f(1,1) = $(1.04^2/1.03)-1 \approx 5.01\%$
- Implied 1-year rate, 1 year from now



PAR RATES & YIELD CURVE THEORIES

Par Rate

- Coupon rate making bond price = par (100)
- Par curve is built from spot curve
- When curve is upward-sloping: spot > par
- Useful for swap pricing (par swap rate)

Yield Curve Shapes

- Normal (upward): long > short rates
- Inverted: short > long (recession signal)
- Flat: rates equal across maturities
- Humped: mid-term rates highest

Expectations	Forward rates are unbiased predictors of future spot rates. Curve shape driven purely by rate expectations.
Liquidity Preference	Long-term investors demand a premium. Curve normally upward-sloping. Forward rate = expected + liquidity premium.
Market Segmentation	Supply and demand within maturity segments determine rates. No spillover between segments.
Preferred Habitat	Investors prefer certain maturities but will move for sufficient premium. Combines segmentation with arbitrage.

Duration & Convexity

LM6, LM12 · Interest rate risk, DV01 & portfolio hedging

DURATION: MEASURING INTEREST RATE SENSITIVITY

Macaulay Duration

Weighted avg

Time to recoup PV of CFs

Modified Duration

%ΔP / Δy

Price % change per 1% yield Δ

Effective Duration

For embedded

options — ΔP / (2×Δy×P)

$$\text{ModDur} = \text{MacDur} / (1 + y/m) \quad | \quad \%\Delta P \approx -\text{ModDur} \times \Delta y$$

Duration Rules

- Duration ↑ as maturity ↑
- Duration ↓ as coupon rate ↑
- Duration ↓ as yield ↑
- Zero-coupon: Duration = Maturity

Dollar Duration (DV01)

- DV01 = ModDur × Price × 0.0001
- $ price change for 1bp yield change
- Portfolio DV01 = sum of bond DV01s
- Used for hedging and position sizing

MACAULAY DURATION: WORKED EXAMPLE

3-Year Bond | 6% Annual Coupon | YTM = 5% | Par = 100

Period (t)	CF	PV of CF @ 5%	Weight (PV/Price)	t × weight
1	6.00	5.714	0.0541	0.054
2	6.00	5.442	0.0515	0.103
3	106.00	91.568	0.8675 (incl FV)	2.602
Total (Price)	**—**	**102.723**	**1.0000**	**MacDur = 2.759 yrs**

$$\text{ModDur} = 2.759 / 1.05 = 2.628$$

$$\%\Delta P \approx -2.628 \times 0.01 = -2.63\%$$

If yields rise by 100bp, price falls approximately 2.63%

PORTFOLIO DURATION & HEDGING

Portfolio Duration

- Weighted average of individual bond durations
- Weight = market value of bond / total portfolio MV
- Portfolio DurP = $\Sigma (w_i \times MD_i)$
- Adding short-duration bonds lowers portfolio risk

$$DurP = \Sigma\ w_i \times MD_i$$

Bond	MV ($M)	Weight	Mod Duration	Contribution
T-Bill	40	40%	0.5	0.20
Corp 5yr	35	35%	4.2	1.47
Corp 10yr	25	25%	7.8	1.95
Portfolio	**100**	**100%**	**—**	**3.62 yrs**

DV01 Hedge: To reduce DV01 by $X, short (or sell) bonds with offsetting DV01. Futures also used.

CONVEXITY: IMPROVING THE DURATION APPROXIMATION

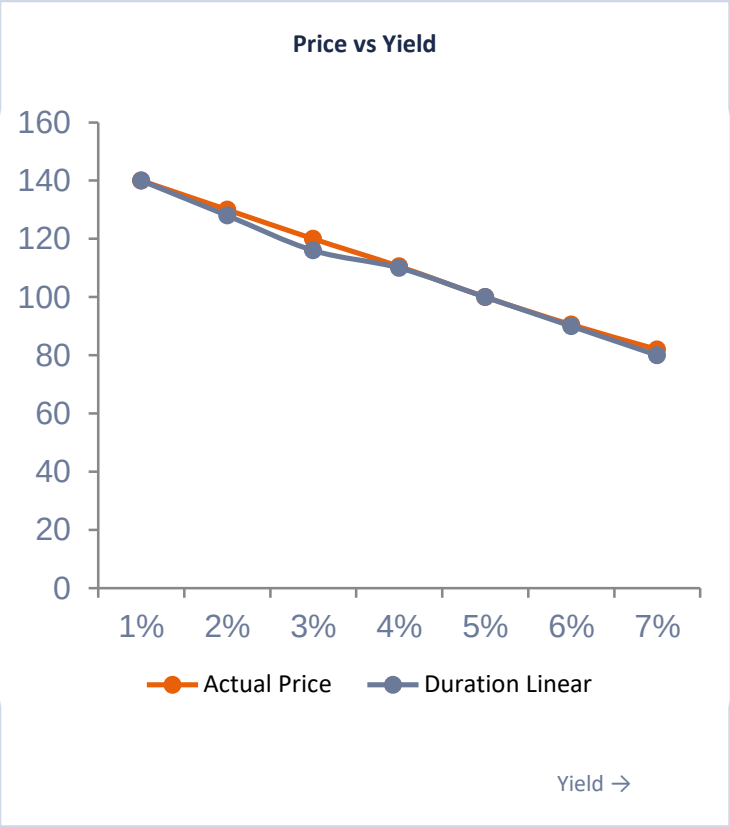
Why Convexity Matters

- Duration is a linear approximation — price-yield is curved
- Convexity measures the curvature of the price-yield relationship
- Bonds with higher convexity: better price performance both up & down
- Convexity is always positive for straight (option-free) bonds

$$\%\Delta P \approx -\text{ModDur} \times \Delta y + \tfrac{1}{2} \times \text{Convexity} \times (\Delta y)^2$$

Convexity Rules

- Higher for longer maturity, lower coupon bonds
- Negative convexity: callables near par (price capped by call)
- MBS have negative convexity due to prepayment risk



Asset-Backed Securities

LM5 · Securitisation, MBS, prepayment risk & tranching

SECURITISATION PROCESS

1

Originator

Bank originates loans (mortgages, auto, credit cards)

2

SPV (Issuer)

Special Purpose Vehicle buys loans from originator (true sale)

3

Structuring

Loans pooled; securities issued in tranches (AAA→equity)

4

Credit Support

Overcollateralisation, excess spread, reserve account, guarantees

5

Investors

Buy ABS tranches matching their risk/return appetite

MORTGAGE-BACKED SECURITIES & PREPAYMENT RISK

Mortgage Pass-Through

- Investors receive pro-rata share of pool payments
- Payments = principal + interest + prepayments
- WAM (weighted avg maturity) changes with prepayments
- PSA benchmark: prepayment speed assumption

CMO Tranches

- Sequential pay: A tranche paid first (shortest duration)
- PAC (Planned Amortisation Class): stable cash flows
- Support/Companion: absorbs variability
- Z-tranche: no cash flows till others paid (long duration)

Prepayment Risk Components

Contraction Risk

Rates ↓ → prepayments accelerate → investor receives principal early at low rates

Extension Risk

Rates ↑ → prepayments slow → investor stuck in low-coupon pool longer

Credit Analysis

LM7, LM15–16 · Ratings, sovereign & corporate credit

CREDIT RISK: COMPONENTS & MEASUREMENT

Probability of Default

PD

Likelihood issuer fails to pay

Loss Given Default

LGD

= 1 − Recovery Rate

Expected Loss

EL = PD × LGD

Annual loss estimate

`Credit Spread ≈ PD × LGD (simplified) | Yield = Risk-free rate + Credit spread + Liquidity premium`

Rating Agencies

- Moody's: Aaa, Aa, A, Baa / Ba, B, Caa, Ca, C
- S&P/Fitch: AAA→D scale
- Investment Grade: BBB−/Baa3 and above
- High Yield (junk): BB+/Ba1 and below

Rating vs Market Spread

- Ratings: backward-looking opinion
- Spreads: market's current view (forward-looking)
- Fallen angels: IG downgraded to HY
- Rising stars: HY upgraded to IG

THE 4 CS OF CREDIT ANALYSIS

Capacity

- Ability to service debt from cash flows
- Leverage: Net Debt/EBITDA, Debt/Equity
- Coverage: EBITDA/Interest expense
- Free cash flow generation consistency

Collateral

- Assets securing the debt
- Seniority: secured vs unsecured
- Recovery rates: senior secured ~70%, senior unsecured ~40%
- Asset quality, liquidity of collateral

Covenants

- Affirmative: maintain insurance, pay taxes
- Negative: no add'l debt, dividend restrictions
- Maintenance: test quarterly (bank debt)
- Incurrence: test only on trigger event

Character

- Management quality & track record
- Shareholder vs bondholder-friendly
- History of meeting obligations
- Strategy: growth vs capital return

SOVEREIGN CREDIT ANALYSIS

Unique Sovereign Features

- Sovereigns can default despite printing own currency
- Local currency debt: lower default risk (can print)
- Foreign currency debt: higher risk (cannot print FX)
- IMF and Paris Club as lenders of last resort

Willingness vs Ability to Pay

- Ability: fiscal space, reserve levels, debt/GDP
- Willingness: political will, investor relations
- Ecuador 2008: chose to default on external debt
- Analysing both is essential for sovereign credit

Key Sovereign Rating Factors

Economic Strength	GDP growth, diversification, per capita income, external trade
Fiscal Strength	Revenue/GDP, deficit, debt/GDP, debt structure (FX vs local)
External Position	Current account, FX reserves vs imports, external debt
Institutional	Rule of law, corruption, central bank independence, political risk
Event Risk	Contingent liabilities, banking sector exposure, geopolitical

CORPORATE CREDIT ANALYSIS

Industry Analysis First

- Cyclical vs non-cyclical (defensive sectors)
- Competitive dynamics: Porter's 5 forces
- Capital intensity & asset tangibility
- Regulatory environment
- Industry credit profile = baseline

Key Credit Metrics

- Leverage: Net Debt / EBITDA (< 2x = low, > 5x = high)
- Coverage: EBIT / Interest expense (> 3x healthy)
- FCF yield: FCF / Total debt
- Debt / Capital ratio
- Return on invested capital (ROIC)

Corporate Credit: Waterfall of Claims

Senior Secured — First lien on specific assets. ~65–80% recovery

Senior Unsecured — Claim on all assets after secured. ~35–50% recovery

Subordinated — Behind senior unsecured. ~15–25% recovery

Equity / Hybrid — Last in priority. Typically ~0% in distress

HIGH YIELD VS INVESTMENT GRADE

Feature	Investment Grade (IG)	High Yield (HY)
Rating	BBB−/Baa3 and above	BB+/Ba1 and below
Typical spread	50–200 bps over govt	300–1,000+ bps over govt
Coupon	Lower, more stable	Higher; compensates for credit risk
Leverage	< 3x Net Debt/EBITDA	> 4–6x Net Debt/EBITDA
Covenant protection	Weak (incurrence only)	Stronger (maintenance covenants)
Main risk driver	Interest rate duration	Credit / default risk
Investor base	Insurance, pension funds	Hedge funds, credit specialists
Correlation with equities	Low–moderate	High (behaves like equity in stress)

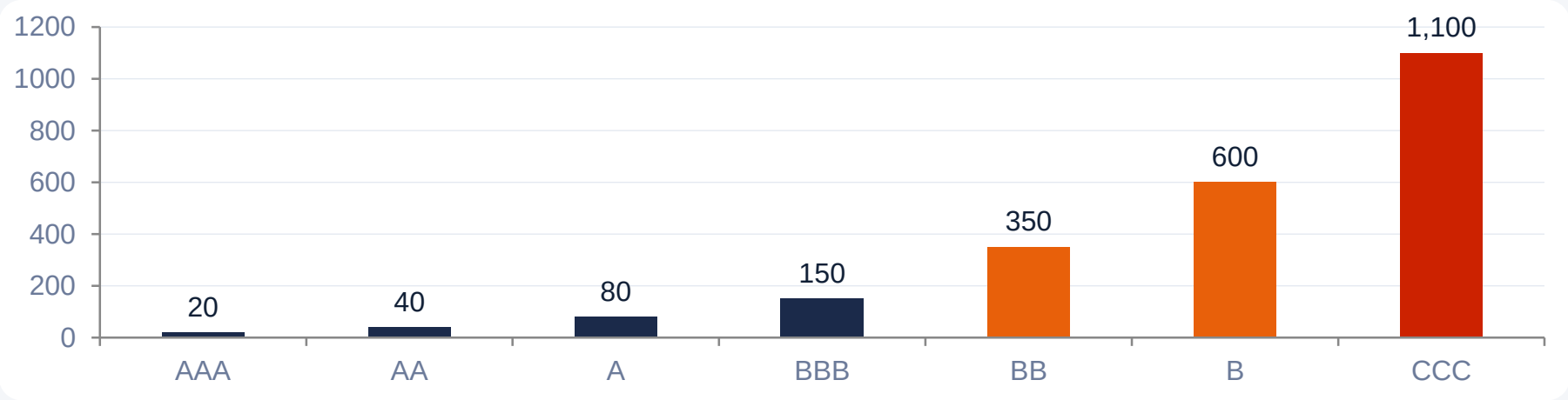
CREDIT SPREAD ANALYSIS

What Drives Credit Spreads?

- PD and LGD expectations (fundamentals)
- Economic cycle: spreads widen in recession
- Liquidity: wider in stress / thin markets
- Supply / demand for credit risk
- Market volatility (VIX correlates with spreads)

Spread Measures

- Nominal spread: YTM bond – YTM benchmark
- G-spread: vs govt curve interpolated
- I-spread: vs swap curve
- Z-spread: constant shift to spot curve
- OAS: Z-spread minus option value



FLOATING-RATE BONDS & INFLATION-LINKED SECURITIES

Floating-Rate Notes (FRN)

- Coupon = reference rate + quoted margin (QM)
- Typical reference: SOFR, EURIBOR
- Duration ≈ time to next reset (very short)
- FRN risks: credit risk of issuer, floor/cap features
- Discount margin (DM): spread vs reference rate given price

Inflation-Linked Bonds (ILBs)

- Principal indexed to CPI; coupons based on indexed principal
- TIPS (USA): most liquid global ILB market
- Real yield = nominal yield − breakeven inflation
- Deflation floor on TIPS: no loss of initial par
- Duration: longer than comparable nominal bonds

FRN Price: if QM = DM → price = par

Real Yield = Nominal − BEI

Key Comparison

- FRNs hedge interest-rate risk; ILBs hedge inflation risk. Both are valuable in diversified fixed income portfolios.

CALLABLE, PUTABLE & CONVERTIBLE BONDS

Callable Bond

- Issuer can redeem at specified price before maturity
- Call protects issuer vs falling rates (refinancing benefit)
- Investor requires higher yield (yield to worst = min yield)
- Negative convexity near par: price capped by call
- OAS removes option value from Z-spread

Putable Bond

- Holder can sell back at specified price
- Put protects investor vs rising rates
- Lower yield than equivalent straight bond
- Positive convexity enhanced by put floor
- Typically putable once or periodically

Value of Options

Bond Type	Yield vs Straight	OAS vs Z-spread	Convexity	Duration
Straight bond	Reference	Equal	Positive	Standard
Callable	Higher (issuer-friendly)	OAS < Z-spread (option costs investor)	Negative at low yields	Shorter (callable = lower)
Putable	Lower (investor-friendly)	OAS > Z-spread (put adds value)	More positive	Shorter at high yields

EUROBONDS & GLOBAL FIXED INCOME MARKETS

Eurobond Market

- Bonds issued outside issuer's home country
- Denominated in foreign currency (e.g. USD Eurobond from EUR issuer)
- Bearer bonds historically; now immobilised
- Less regulatory oversight; attractive for issuers

Global Bond

- Issued simultaneously in domestic + Euro markets
- Higher liquidity and investor base
- Requires multi-jurisdiction registration
- Tight pricing due to demand breadth

Market	Size (approx.)	Key Issuers
US Treasury	$28 trillion	US Federal Government
Euro Government	€12 trillion	EZ sovereign issuers
US Corporate IG	$6 trillion	Investment grade US corps
US High Yield	$1.5 trillion	Sub-IG US corps
EM Hard Currency	$2.5 trillion	EM sovereigns & corps

INTEREST RATE RISK: FULL FRAMEWORK

Sources of Interest Rate Risk

- Level risk: parallel shift in yield curve
- Slope risk: flattening / steepening
- Curvature risk: butterfly shifts
- Volatility risk: affects option-embedded bonds
- Spread duration: spread change on credit bonds

Risk Measures Summary

- Macaulay Duration: weighted time to CFs
- Modified Duration: %ΔP per %Δyield
- Effective Duration: for optionality
- Key Rate Duration: single tenor sensitivity
- DV01: $ change per bp change in yield

$$\text{Full Price } \Delta \approx -MD \times P \times \Delta y + \tfrac{1}{2} \times \text{Convexity} \times P \times (\Delta y)^2$$

Reinvestment Risk

- If yields fall after purchase, reinvested coupons earn less
- Macaulay duration = horizon where both risks offset

Immunisation

- Duration-match assets to liability horizon
- Single-period: set MacDur = investment horizon

CDOS, CLOS & STRUCTURED CREDIT

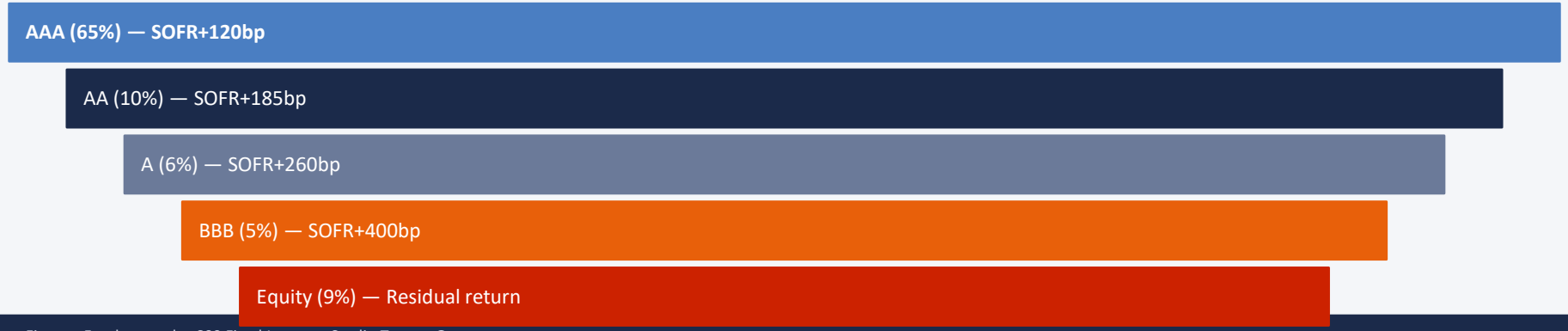
Collateralised Debt Obligations (CDO)

- Pool of bonds/loans tranched into AAA→equity
- Cash flow CDO: backed by actual assets
- Synthetic CDO: backed by CDS on reference assets
- CDO^2 (re-securitisation of CDO tranches)
- Pre-GFC proliferation led to systemic risk 2007–08

CLO Specifics

- Backed by leveraged loans (not bonds)
- Active manager selects/trades loan pool
- AAA tranches: ~60–65% of structure
- Equity tranche: ~9–10%; residual return

Waterfall & Credit Enhancement



CONVERTIBLE BONDS

Convertible Bond Features

- Holder may convert into issuer's equity at conversion price
- Conversion ratio = par value / conversion price
- Conversion value = conversion ratio × stock price
- Minimum value = max(straight value, conversion value)
- Busted convertible: equity option deep out-of-the-money

Conversion Metrics

- Conversion price: pre-set strike price
- Market conversion price = Bond / Conversion ratio
- Conversion premium = Mkt conv. price − Stock price
- Premium over straight: cost of equity option

$$\text{Conversion Value} = \text{Conversion Ratio} \times \text{Stock Price}$$

$$\text{Min Value} = \max(SV, CV)$$

Investment Characteristics

- Hybrid instrument: bond floor (downside protection) + equity call (upside capture)
- Busted convertibles trade like straight bonds; equity-sensitive convertibles trade like stock with income

FIXED INCOME PORTFOLIO CONSTRUCTION

Portfolio Objectives

- Liability matching (ALM): pension, insurance
- Return maximisation within risk budget
- Income generation: coupon cash flows
- Capital preservation: short duration, high grade
- Benchmark relative: tracking error management

Strategies

- Passive indexing: replicate benchmark
- Duration management: active duration tilts
- Sector rotation: shift allocation by cycle
- Credit tilts: overweight IG or HY vs benchmark
- Yield curve positioning: bullet, barbell, ladder

Yield Curve Structures

Bullet	Barbell	Ladder
Concentrated around one maturity. Maximises curve bet exposure.	Short + long maturities only. More convexity than bullet at same duration.	Equally spaced maturities. Reinvestment diversification, predictable income.

FIXED INCOME — FORMULA REFERENCE SHEET

Bond Price

$$P = \Sigma\ CF_t/(1+y)^t$$

Accrued Interest

$$AI = C \times t/T$$

Full Price

$$PFull = PClean + AI$$

Current Yield

$$CY = Annual\ Coupon\ /\ P$$

Macaulay Duration

$$MacDur = \Sigma\ [t \times PV(CF_t)]\ /\ P$$

Modified Duration

$$ModDur = MacDur\ /\ (1+y/m)$$

%ΔP (Duration)

$$\%\Delta P \approx -ModDur \times \Delta y$$

%ΔP (Duration+Convex)

$$\%\Delta P \approx -MD \times \Delta y + \tfrac{1}{2} \times C \times (\Delta y)^2$$

Forward Rate

$$(1+S_n)^n = (1+S_m)^m \times (1+f_{m,n})^{n-m}$$

Expected Loss

$$EL = PD \times LGD$$

Portfolio Duration

$$DurP = \Sigma\ w_i \times MD_i$$

Conversion Value

$$CV = Ratio \times Stock\ Price$$